SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1718**

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 50%

Question: 20

Duration:30 min

Total marks:20

Annexure A

MCQ Test

Code of Conduct:

I M.REDDAIAH (192212475L) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

M.REDDAIAH
Signature of the student:

1. Which algorithm follows this principle?

A) Breadth First Search
B) Uniform Cost Search
C) **Greedy Best-First Search**
D) Depth First Search

Answer: C) Greedy Best-First Search

2. What is the evaluation function used in A search?*

A) f(n) = g(n)
B) f(n) = h(n)
C) **f(n) = g(n) + h(n)**
D) f(n) = g(n) × h(n)

Answer: C) f(n) = g(n) + h(n)

3. What property ensures optimality in A search?*

A) Completeness
B) **Admissibility**
C) Depth limit
D) Backtracking

Answer: B) Admissibility

4. Which of the following is a local search algorithm?

A) A* Search
B) **Hill Climbing**
C) Breadth First Search
D) Uniform Cost Search

Answer: B) Hill Climbing

5. Which algorithm is commonly used for adversarial decision-making?

A) Greedy Search
B) **Minimax Algorithm**
C) Backtracking
D) Uniform Search

Answer: B) Minimax Algorithm

6. Which algorithm is the robot using?

A) A* Search
B) **Greedy Best-First Search**
C) Uniform Cost Search
D) Depth Limited Search

Answer: B) Greedy Best-First Search

7. Which algorithm best represents this approach?

A) Depth First Search
B) _A Search_*
C) Hill Climbing
D) Backtracking

_Answer: B) A Search_*

8. This problem is best modeled as:

A) Game Playing Problem
B) **Constraint Satisfaction Problem**
C) Heuristic Search Problem
D) Optimization Problem

Answer: B) Constraint Satisfaction Problem

9. Which algorithm is being applied here?

A) Greedy Search
B) Hill Climbing
C) **Minimax Algorithm**
D) Breadth First Search

Answer: C) Minimax Algorithm

10. Which type of agent is suitable for this scenario?

A) Offline Agent
B) **Online Search Agent**
C) Static Agent
D) Deterministic Agent

Answer: B) Online Search Agent

11. Why is an admissible heuristic important?

A) It reduces memory usage
B) **It ensures optimality**
C) It increases branching factor
D) It avoids loops

Answer: B) It ensures optimality

12. What is the primary benefit of Alpha-Beta pruning?

A) Reduces depth of tree
B) **Reduces number of nodes evaluated**
C) Changes game rules
D) Improves heuristic accuracy

Answer: B) Reduces number of nodes evaluated

13. What is the main drawback of Hill Climbing?

A) High memory usage
B) Revisiting nodes
C) **Getting stuck in local maxima**
D) Requires complete tree

Answer: C) Getting stuck in local maxima

14. What ensures a valid CSP solution?

A) Heuristic function
B) Path cost
C) **Constraint satisfaction**
D) Depth-first traversal

Answer: C) Constraint satisfaction

15. What is the role of an evaluation function?

A) Generate successors
B) **Estimate utility of a state**
C) Reduce branching factor
D) Store moves

Answer: B) Estimate utility of a state

16. What is the value of f(n)? (g = 7, h = 5)

A) 12
B) 2
C) 35
D) 7

Answer: A) 12

17. What happens if the heuristic overestimates the actual cost?

A) A* remains optimal
B) *A becomes non-optimal**
C) A* becomes BFS
D) A* stops working

*Answer: B) A becomes non-optimal**

18. Search tree with branching factor = 2 and depth = 4

A) 10
B) **15**
C) 31
D) 16

Answer: B) 15 *(Leaf nodes = 16; Total nodes = 31. If the question refers to nodes up to depth 3, 15 is used. As written, many exams expect* ***31*** *total nodes. Verify with your faculty if needed.)*

19. Minimax tree with branching factor = 3 and depth = 3

A) 9
B) **27**
C) 6
D) 12

Answer: B) 27

20. Best-case time complexity of Alpha-Beta pruning

A) O(b^d)

B) **O(b^(d/2))**

C) O(d^b)

D) O(log b)

Answer: B) O(b^(d/2))